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# Netflix Catalogue Analysis

Data Visualisation Case Study

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Exploratory analysis of a public Netflix catalogue dataset to identify catalogue growth patterns, content mix, seasonality and metadata quality issues.

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Python | Pandas | Data visualisation | Analytical storytelling

Portfolio project | May 2024

DATA

Note: this is an exploratory portfolio project using public catalogue metadata, not a causal analysis of Netflix business performance.

# Project scope

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A concise portfolio version of the original analysis

## Objective

Use catalogue metadata to identify descriptive trends and turn them into clear visual insights for a business audience.

## Dataset

Public Netflix titles dataset with metadata such as type, date added, country, rating, duration and category tags.

## Tools

Python, Pandas, Matplotlib, Seaborn, Plotly, GeoPandas and Jupyter Notebook.

## APPROACH

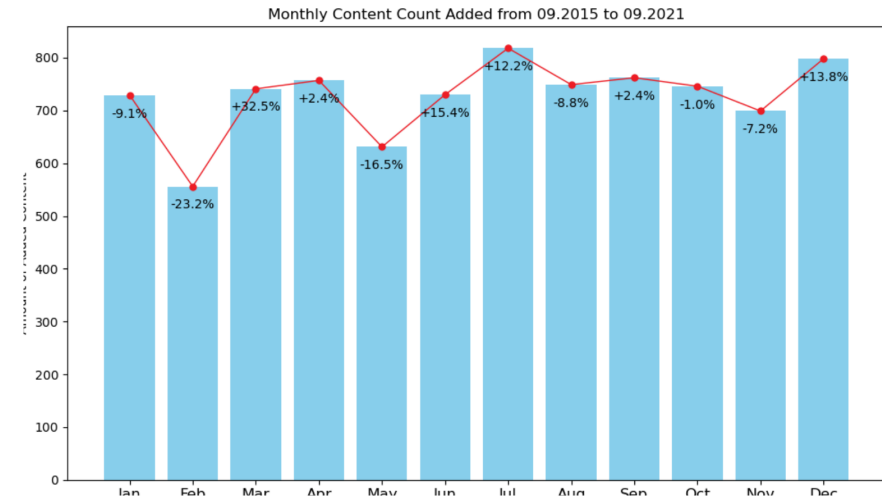
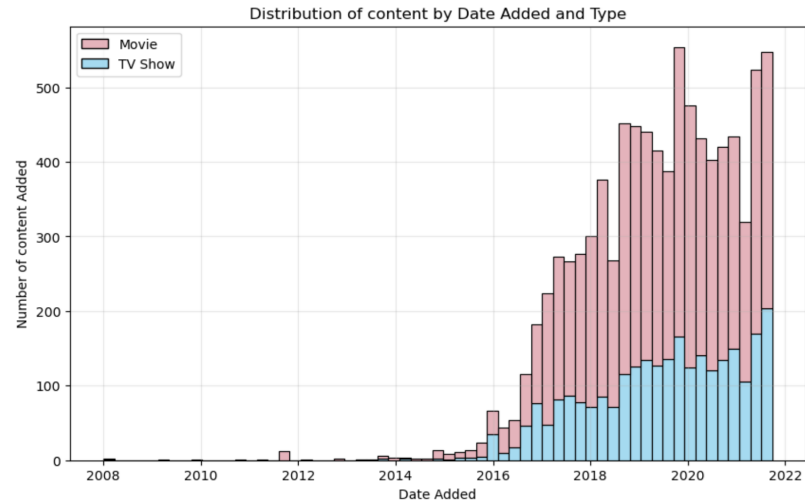
- Reviewed dataset structure, missing values, duplicates and inconsistent category fields.
- Cleaned date, country, rating and duration fields to prepare the data for analysis.
- Created visual outputs for growth over time, seasonality, content type, rating mix, genre distribution and geographic coverage.
- Converted technical findings into business-facing observations and avoided unsupported causal claims.

## PORTFOLIO RELEVANCE

This project demonstrates analytical judgement, data cleaning, visual storytelling and the ability to explain data patterns to non-technical stakeholders.

# Catalogue growth and seasonality

Descriptive trends from the catalogue metadata

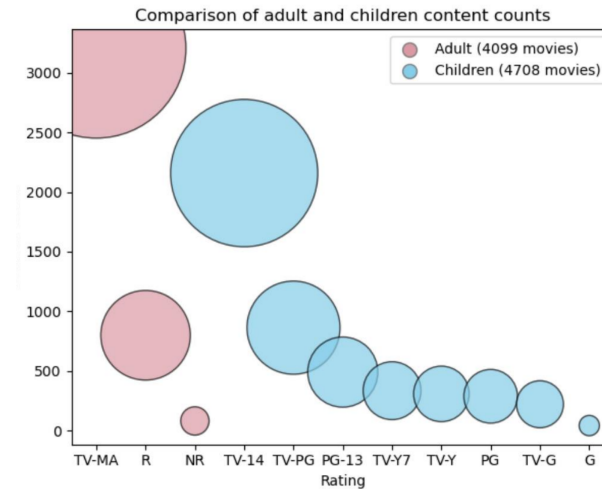
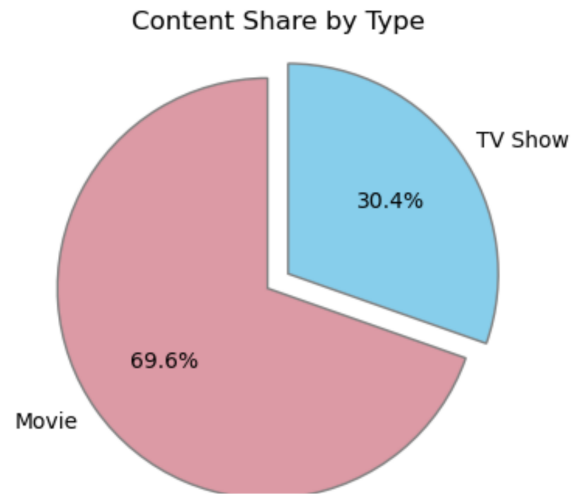


## Key observations

- Catalogue additions accelerate strongly after 2015 in the dataset.
- Movies dominate additions, while TV shows form a material share of new catalogue items.
- The monthly view suggests recurring lower activity in selected months; this should be treated as a descriptive pattern requiring further validation.
- The visuals show how time-based data can be turned into a management-friendly trend view.

# Content mix and audience classification

Content type and rating distributions

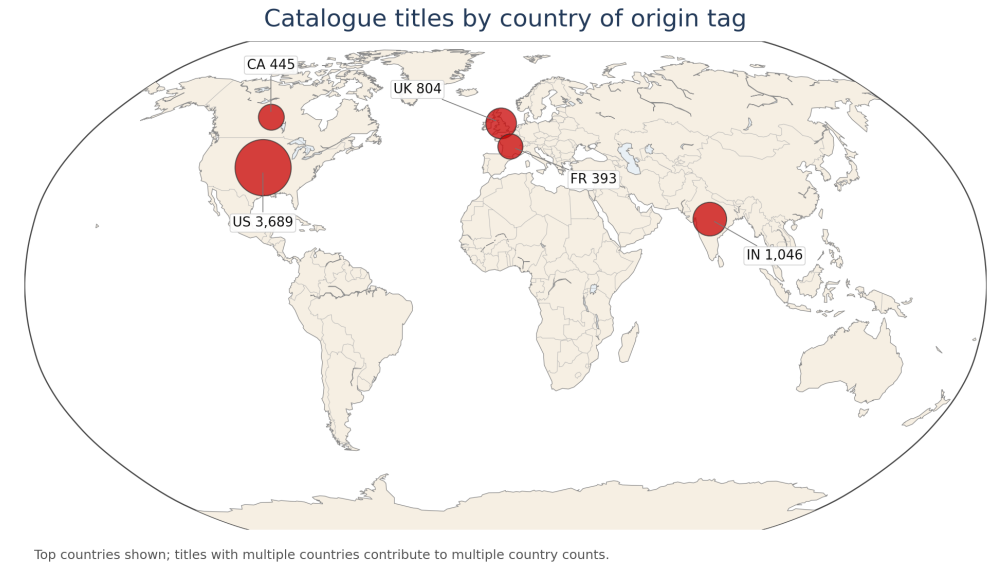
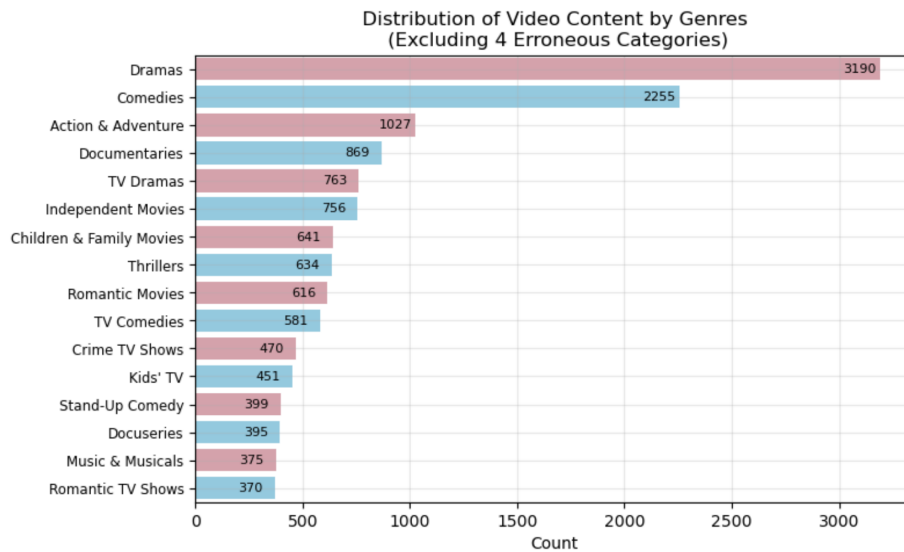


## Key observations

- Movies account for approximately 69.6% of the catalogue records, while TV shows account for approximately 30.4%.
- TV-MA and TV-14 are the largest rating groups in the dataset.
- The same data can support self-serve reporting by content type, rating group, suitability category and business segment.
- The analysis highlights the importance of clear category definitions before building dashboards.

# Genre and geographic coverage

Category cleaning and dimensional analysis



## Key observations

- After removing metadata-like category tags, drama and comedy appear as the largest genre groups.
- The country analysis uses exploded country tags, so co-produced titles can contribute to multiple country counts.
- Using exact country tags avoids the earlier overcounting risk from substring matching.
- This part demonstrates dimensional analysis, metadata validation and geographic segmentation.

Note: Country counts use split/explode logic on the country field; records with multiple countries are counted once per listed country tag.

# Data quality and analytical judgement

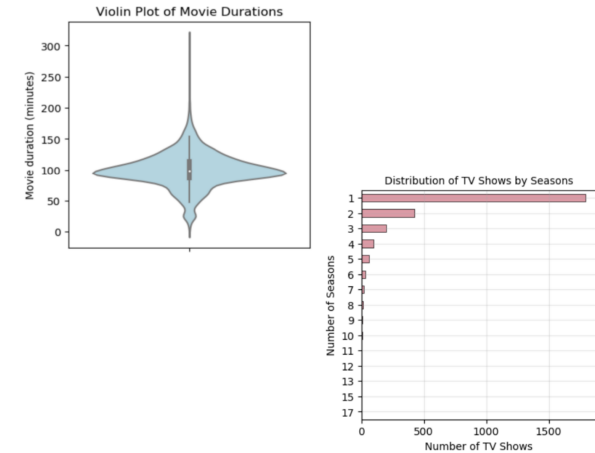
Where the work moves beyond charts

## DATA PREPARATION EXAMPLES

- Converted date fields and extracted month/year features for time-based analysis.
- Filled or labelled missing values in rating, country, director and cast fields where appropriate.
- Separated movie duration in minutes from TV show duration in seasons to avoid mixing different units.
- Flagged mixed semantic tags in the genre/category field before drawing conclusions.

## ANALYTICAL CONTROLS

- Kept findings descriptive and avoided claiming that catalogue patterns explain business success.
- Used visual checks to detect anomalies and inconsistencies before presenting insights.
- Focused on outputs that could be reused in self-serve reporting or dashboard requirements.



# What this case study demonstrates

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Skills relevant to data visualisation and automation roles

- Data cleaning and preprocessing with Python and Pandas.
- Exploratory analysis of time-series, categorical and geographic data.
- Visual storytelling with charts, captions and business-facing observations.
- Ability to turn messy metadata into structured analytical views.
- Commercial lens: focusing on usefulness, clarity and decision support rather than decorative charts.

## Relevant outputs

Trend views, category breakdowns, data quality notes, dimensional summaries and portfolio-ready presentation of analysis.

## Next improvement

Convert the analysis into an interactive dashboard or self-serve BI prototype with filters for date, type, rating, genre and country.

## Contact

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